

SRI VENKATESWARA COLLEGE OF ENGINEERING

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Pennalur, Sriperumbudur Tk, Tamil Nadu - 602 117

OFFICIAL TECHNICAL & FUNCTIONAL DOCUMENTATION

Autonomous No-Dues Smart ERP System

A Comprehensive 6-Stage Digital Clearance Workflow, Relational MySQL Architecture, Role-Based Institutional Portals, Cryptographic Certificate Engine & Progressive Web Application (PWA).

ARCHITECTURE React 18 • Express • MySQL 8	CLEARANCE ENGINE 6 Autonomous Stages	SECURITY & AUDITING GPS & IP Geolocation Logs
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System: SVCE No-Dues Enterprise Edition v2.4
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1. Executive Summary & Problem Formulation

1.1 Institutional Background & Traditional Bottlenecks

In higher education institutions, graduating students and candidates sitting for end-semester examinations are mandated to obtain institutional "No-Dues Clearance" across multiple autonomous academic, administrative, and operational bodies. Historically at engineering institutions, this clearance relied upon physical paper cards requiring in-person signatures from multiple geographically separated departments:

- **Department Library:** Physical verification of accession cards and returned textbooks.
- **Department Placement Cell (DPC):** Manual inspection of company offer letters or higher education acceptance letters.
- **Central Library OPAC Desk:** In-person verification across multiple library stacks and fine reconciliation.
- **Faculty Advisor (FA):** Verification of academic eligibility, minimum 75% attendance threshold, and hall ticket generation.
- **Finance & Accounts Office:** Tuition fee audit, bus transport fees, laboratory caution deposits, and hostel billing reconciliation.
- **Head of Department (HOD):** Manual cross-checking and final ink endorsement on paper certificates.

Traditional Process Deficiencies: Physical clearances caused massive administrative congestion at semester-end, students standing in queues across multiple campus blocks, misplaced physical records, lack of real-time auditability, and delayed issuance of digital Hall Tickets and Degree Certificates.

1.2 The Smart No-Dues Solution

The **SVCE No-Dues Smart ERP System** transforms this entire institutional workflow into a 100% paperless, fully automated, sequential 6-stage digital clearance pipeline. Built with enterprise-grade security, live device footprinting, precise geographic location tracking, and an installable Progressive Web Application (PWA), the system delivers complete visibility and real-time execution across 7 distinct institutional user roles.

⚡ 95% Reduction in Clearance Time

Clearance completion reduced from 4–7 days to under 4 hours via automated verification gates and real-time notifications.

🔒 Cryptographic Tamper-Proofing

Digital clearance certificates embedded with unique SHA-256 validation hashes, dynamic QR codes, and institutional watermarks.

📱 Multi-Platform PWA Support

Works as a responsive web app and installable desktop/mobile application with service worker offline caching.

📍 Authentic Device & Location Auditing

Real-time GPS coordinates and IP-based device tracking logged on all logins, ensuring strict institutional accountability.

2. System Architecture & Technology Stack

2.1 Full-Stack Technical Architecture

The system follows a decoupled Model-View-Controller (MVC) and Component-Driven Client-Server Architecture, engineered for high concurrency, zero data loss, and seamless responsive execution.

LAYER	TECHNOLOGIES & LIBRARIES	INSTITUTIONAL RESPONSIBILITY & ROLE
Frontend Client	React 18, Vite 5, Vanilla CSS, Tailwind Utility, Lucide React	Component-based single-page application (SPA), dynamic role-based dashboards, modal management, custom alerts, QR rendering, and print-ready CSS engines.
Mobile & PWA Engine	Vite PWA Plugin, Workbox, Web Manifest	Service Worker background precaching of 34 production assets, install prompts across iOS/Android/Windows/macOS, standalone display mode, and offline resilience.
Backend API Server	Node.js (LTS), Express.js, SSE (EventSource)	RESTful micro-endpoints, role-based JWT auth verification, multipart file upload handling, device/IP extraction, and server-sent event push notifications.
Database Tier	MySQL 8.0, mysql2/promise, Connection Pool	ACID-compliant relational database, foreign key cascades, automatic self-healing MySQL server detection, transactional clearance execution, and zero SQLite.
Security & Auth	JWT (JsonWebToken), bcryptjs, CORS	Cryptographic password hashing (10 salt rounds), 12-hour signed session tokens, role authorization middleware, and parameterized SQL query injection immunity.
Location & Geocoding	HTML5 Geolocation, Reverse Geocoder, IP-Geo Lookup	High-accuracy hardware GPS coordinates, reverse geocoding to city/neighborhood, URL-safe header encoding, and IP location fallback.

2.2 Relational Database Schema Design (Pure MySQL)

The project utilizes a relational MySQL schema (`svce_nodues`), eliminating all file-based SQLite locks to ensure enterprise reliability.

TABLE NAME	PRIMARY KEY	KEY RELATIONSHIPS	FUNCTIONAL PURPOSE
users	id (INT AUTO)	Referenced by all role profiles	Central authentication store containing usernames, bcrypt passwords, emails, and roles.
students	id (INT AUTO)	FK: user_id → users.id	Full academic record: Register No, Name, Dept, Batch, Semester, Advisor Emp ID, Career status.
nodues_requests	id (INT AUTO)	FK: student_id → students.id	Clearance master record: submission date, career category, overall status, progress %, certificate hash.
nodues_stages	id (INT AUTO)	FK: request_id → nodues_requests.id	The 6 individual stage records storing stage order, status, remarks, updated_by, and timestamp.
nodues_audit_logs	id (INT AUTO)	FK: request_id → nodues_requests.id	Permanent departmental action trail tracking re-submissions, proof URLs, and approvals.
books	id (INT AUTO)	Referenced by borrow_records	Library catalog: ISBN, Title, Author, Department, Total Copies, Available Copies.
borrow_records	id (INT AUTO)	FK: student_id, book_id	Circulation registry tracking issue dates, due dates, return status, and accrued overdue fines.
complaints	id (INT AUTO)	FK: student_id → students.id	Grievance desk tickets with departmental routing, status (Open/Resolved), and admin responses.
announcements	id (INT AUTO)	created_by (VARCHAR)	Institutional broadcast notifications categorized by priority (General, Clearance, Exam, Urgent).
system_audit_logs	id (INT AUTO)	FK: user_id → users.id	Enterprise login session registry recording timestamp, operator, device name, type, IP, and GPS location.

3. The 6-Stage Autonomous Clearance Pipeline

Clearance requests progress sequentially through six institutional gates. A request cannot proceed to subsequent stages until prerequisite departmental gates are endorsed. Furthermore, if a stage is placed on **Hold** or **Rejected**, the student can submit a targeted re-submission with proof documents directly to that specific department without repeating previously approved stages.

Stage 1: Department Library Verification

DEPARTMENTAL GATE

Conducted by the Information Technology Department Library In-Charge. Verifies that all departmental books, reference project manuals, and technical journals issued to the student have been returned in undamaged condition.

Verification Criteria: Active borrowed book count = 0 • Department overdue fines = ₹0 • Physical textbook inspection verified.

Stage 2: Department Placement Coordinator (DPC)

CAREER GATE

Conducted by the IT Department Placement Coordinator. Validates the graduating 4th-year student's career outcome. The student must declare one of three pathways and provide verified proof documents:

Pathway A (Campus Placement): Company Name, Designation, CTC (LPA), Offer Letter PDF Upload.
Pathway B (Higher Studies): University Name, Degree Program, Country, Admission Letter/Scorecard.
Pathway C (Entrepreneurship/Competitive Exam): Incorporation proof or official exam registration.

Stage 3: Central Library Institutional Clearance

CAMPUS-WIDE GATE

Conducted by the Central Library Clearance Officer. Scans the institutional OPAC database across all central stacks, digital library media, and inter-library loan records.

Verification Criteria: Zero pending central library books • Complete settlement of overdue circulation fines • ID card barcode scan confirmation.

Stage 4: Faculty Advisor (FA) Academic Endorsement

ACADEMIC & EXAM GATE

Conducted by the student's designated Faculty Advisor (e.g., 3rd/4th Year IT Section Advisor). Performs a comprehensive 4-point academic audit before granting clearance and issuing digital Hall Ticket eligibility.

4-Point Audit Criteria: (1) Minimum 75% attendance threshold • (2) Continuous Internal Assessment (CIA) marks entry verified • (3) Laboratory record notebook completion • (4) Satisfactory student conduct and disciplinary record.

Stage 5: Finance Section & Accounts Audit

FINANCIAL GATE

Conducted by the Finance Clearance Officer in the Central Administrative Block. Validates all monetary transactions across the student's entire academic tenure.

Financial Reconciliation: Tuition fee payment ledger • College bus/transport fee audit • Laboratory consumables/breakage charges • Examination fees • Caution deposit adjustments.

Stage 6: Head of Department (HOD) Final Authorization

EXECUTIVE SEAL GATE

Conducted exclusively by the Head of the Department (e.g., Dr. V. Vidhya, HOD-IT). Reviews the aggregated approvals across all prior 5 stages, conducts executive review, and applies the official institutional digital seal.

Executive Trigger: 100% completion of Stages 1–5 • Final sign-off logs recorded • Cryptographic certificate hash generated • Digital No-Dues Certificate immediately unlocked for student download.

4. Role-Based Dashboards & Module Functionality

4.1 Student Portal (The Student Clearance Experience)

The student interface is engineered to be intuitive, clean, and completely student-centric. In strict adherence to administrative security protocols, internal staff audit logs are completely hidden from the student view, replaced by a visual 6-stage interactive timeline.

COMPONENT / TAB	FUNCTIONAL CAPABILITIES & IMPLEMENTATION
ERP Dashboard (Overview)	Displays summary metric cards (Active borrowed books, overdue fines, Department Library status, Overall No-Dues Progress %). Displays comprehensive student academic profile and assigned Faculty Advisor details with direct phone/email contact.
No-Dues Tracker	Interactive visual tracking pipeline with progress bar (0% to 100%), active status badges (Pending, In Progress, Approved, Rejected, Hold), official officer remarks for each department, and targeted <i>Resubmit Request</i> modal with proof document attachment upload.
Digital Certificate	Unlocked automatically once Stage 6 (HOD) is approved. Features institutional crest, student portrait, academic credentials, QR code verification link, cryptographic hash, and direct <i>Print / Download PDF</i> layout formatted for A4 printing.
Complaint Desk	Allows students to file support tickets regarding clearance delays, incorrect fine billing, or book discrepancy. Tickets are categorized by department and priority, with live resolution tracking.
Announcements Feed	Campus broadcast board displaying critical institutional notices, clearance deadlines, and circulars issued by HOD or Dean.
Settings & Security	Allows students to update mobile phone contact info, profile photo URL, and change account password with bcrypt re-hashing.

4.2 Department Library Staff Portal

Serves the IT Department Library In-Charge. Features two primary sub-systems: the **No-Dues Clearance Desk** and the **Book Circulation Inventory**.

- **No-Dues Clearance Desk:** Real-time queue of all departmental clearance applications. Searchable by student register number, name, or year. Library officers can endorse Stage 1 with one click, or place on Hold with official remarks detailing missing book titles or overdue fines.
- **Book Inventory Management:** Complete catalog of technical books, project monographs, and reference manuals. Officers can add new titles, update copy counts, and search by ISBN.
- **Student Circulation Directory:** Table listing students with active book borrowings, issue dates, due dates, and calculated daily overdue fines. Officers can record book returns and generate clearance receipts.
- **Login Activity Audit Trail:** Role-scoped login records showing exact login timestamps, device type, and precise GPS location.

4.3 Department Placement Coordinator (DPC) Portal

Specifically oversees Stage 2 clearance for final-year students. Features a high-speed verification queue where placement officers review career proofs:

- **Offer Letter Inspection:** Inspects company name, job role, CTC (LPA), and uploaded offer letter PDFs for placed candidates.
- **Higher Education Verification:** Validates foreign and domestic university admission letters, GATE/GRE/CAT test scorecards.
- **One-Click Endorsement:** Approves Stage 2 with official placement notes or rejects with specific deficiency notices requesting re-submission of verified documentation.

4.4 Central Library Staff Portal

Oversees campus-wide Stage 3 clearances across the central institutional library infrastructure.

- **Institutional OPAC Scrutiny:** Automatically queries student register number against central library circulation databases.
- **Central Dues Ledger:** Inspects and collects central overdue book penalties and non-returned book charges.
- **Exportable Clearance Ledger:** One-click export of daily or term-wide library clearance logs in CSV and JSON formats for institutional archives.

4.5 Faculty Advisor (FA) Portal

Equips faculty advisors with a 360-degree academic and disciplinary command center for their assigned student cohort:

- **Advisee Roster & Analytics:** Visual cards summarizing total advisees, percentage cleared, attendance health, and pending approvals.
- **Stage 4 Approvals Queue:** Reviews student CIA test performance, attendance percentage, and conduct records.
- **Digital Hall Ticket Clearance:** Directly authorizes examination hall ticket permissions once academic criteria are satisfied.
- **Individual Student Deep-Dive:** Inspects student borrow history, active clearance status across prior departments, and disciplinary notes.

4.6 Finance Section Portal

Oversees Stage 5 institutional fee audits and financial reconciliation:

- **Institutional Fee Ledger:** Categorizes tuition fees, examination fees, bus transport charges, and hostel room rents.
- **Fee Clearance Verification:** Validates bank challans, online ERP payment transactions, and scholarship fee waivers.
- **Financial Audit Trail:** Maintains permanent immutable records of financial clearance approvals with operator signatures.

4.7 Head of Department (HOD) Executive Portal

The highest authority portal within the department. Provides comprehensive oversight across all 6 stages and institutional staff:

- **Executive Analytics Dashboard:** Live counters displaying total departmental applications, completed clearances, clearance completion rate (%), and stage bottleneck distribution charts.
- **Stage 6 Final Authorization Desk:** Reviews student records that have achieved 100% approval across Stages 1 through 5. Allows individual or batch final approval, unlocking digital degree certificates immediately.
- **Master Application Registry:** Search, filter, and inspect any student application across any stage, department, or academic year.
- **Department-Wide Login Audit Oversight:** Executive inspection of all login events across all staff roles (HOD, FA, Library, DPC, Finance) with device fingerprints and precise geographic locations.

5. Advanced Engineering Highlights & Security Features

5.1 Progressive Web Application (PWA) Architecture

The application is engineered as a modern Progressive Web App, enabling zero-install access via desktop browsers and native app installation on mobile and desktop platforms.

- **Service Worker Caching (Workbox):** Automatically precaches 34 core assets (HTML, CSS, JS bundles, icons, and SVCE logos) for instantaneous sub-second loading on repeat visits.
- **Web App Manifest:** Complete `manifest.webmanifest` defining `standalone` display mode, brand theme colors (`#1e3a8a` , `#0f172a`), 192×192 and 512×512 app icons, maskable Android icons, and desktop/mobile promotional screenshots.
- **Native Install Banners:** Custom prompt modals and toolbar buttons alerting users on Chrome, Edge, Safari, and Android that the ERP portal can be installed to their home screen or desktop taskbar.

5.2 Authentic Login Auditing with Real Precise Geolocation

Security is maintained through real-time device fingerprinting and precise location auditing. In strict compliance with operational requirements, all fake or hardcoded mock locations have been completely eliminated.

- **Hardware GPS & Wi-Fi Triangulation:** Leverages the browser `navigator.geolocation` API with `enableHighAccuracy: true` to obtain real hardware GPS coordinates (latitude and longitude down to 4 decimal places).
- **Reverse Geocoding:** Maps coordinates into actual city and neighborhood names (e.g., `Chennai, Tamil Nadu, India (13.0878° N, 80.2785° E)`).
- **Public IP Geolocation Fallback:** If GPS is denied or offline, resolves real-time public IP location dynamically.
- **URL-Safe Header Encoding:** Device and location headers are safely encoded to ensure byte compliance with HTTP/1.1 protocols.
- **Role-Segregated Privacy:** Faculty Advisors, Library Staff, and Finance officers only see their own respective login records, while HOD maintains departmental executive oversight.

5.3 Digital Certificate Engine with Cryptographic QR Verification

Certificate Authenticity Architecture: Upon Stage 6 HOD approval, the system generates a unique 16-character cryptographic verification hash (e.g., `SVCE-2026-IT-98F4A2B1`) recorded in the database.

The digital certificate features:

- Institutional header with SVCE crest and Autonomous Anna University endorsement.
- High-resolution student portrait, register number, department, and academic batch.
- Breakdown of all 6 clearance approvals with certifying officer employee IDs and timestamps.
- Dynamic QR Code embedding the live public verification URL. When scanned by external verification bodies or employers, it immediately displays the authentic clearance verification ledger.
- Watermarked security background and print-optimized CSS for flawless export to PDF or physical color printing.

6. Installation, Configuration & Operational Guide

6.1 Prerequisites & System Requirements

- **Node.js:** v18.x or v20.x LTS or higher
- **MySQL Server:** v8.0 or v8.4 Community Edition (running on `localhost:3306`)
- **Web Browser:** Google Chrome 110+, Microsoft Edge 110+, Mozilla Firefox 115+, or Apple Safari 16+
- **Package Manager:** npm v9.x or v10.x

6.2 Step-by-Step Installation Commands

1. Backend Setup & Configuration

```
# Navigate to server directory
cd server

# Install server dependencies (Pure MySQL2, Express, bcryptjs, jsonwebtoken)
npm install

# Configure environment variables (.env)
PORT=5000
DB_HOST=localhost
DB_USER=root
DB_PASSWORD=root
DB_NAME=svce_nodues
JWT_SECRET=svce_nodues_secret_key_2026_secure

# Start the server (auto-detects and connects to MySQL)
npm run dev
```

2. Frontend Client Setup

```
# Navigate to client directory
cd client

# Install client dependencies
npm install

# Start Vite development server
npm run dev
# Running on http://localhost:3000

# Build production bundle with PWA service worker
npm run build
```

6.3 Built-in Demonstration Accounts & Credentials

The system comes pre-initialized with representative test accounts across all 7 institutional roles (all accounts use password: `password123`):

ROLE	OFFICER / USER NAME	REGISTER NO. / EMPLOYEE ID	DESIGNATION / NOTES
Head of Department	Dr. V. Vidhya	EMP-HOD-IT-01	HOD - Information Technology (Stage 6 Authorizer)
Faculty Advisor	V Praveenkumar	EMP-FA-IT-01	FA - 3rd/4th Year IT-B Cohort (Stage 4 Authorizer)
Faculty Advisor	S. Kavishree	EMP-FA-IT-04	FA - 3rd Year IT-A Cohort
Placement (DPC)	Dr. R. Placement Coordinator	EMP-DPC-IT-01	IT Department Placement In-Charge (Stage 2 Authorizer)
Department Library	Sivakumar E	EMP-LIB-IT-01	IT Dept Library In-Charge (Stage 1 Authorizer)
Central Library	Mohan Kumar S	EMP-MLIB-IT-01	Central Library Clearance Officer (Stage 3 Authorizer)
Finance Section	Gurusamy M	EMP-FIN-IT-01	Accounts & Clearance Officer (Stage 5 Authorizer)
Student (Placed)	Aadhityan K	IT2024001	4th Year IT-A (Placed at Zoho Corporation - 8.5 LPA)
Student (Higher Studies)	Bhavani S	IT2024002	4th Year IT-A (Admitted to MS Program)
Student (3rd Year)	Sharvesh C M	IT2025084	3rd Year IT-B (Academic Candidate)

Quick Login Feature: On the Login Page, users can click any of the *Demo Account Quick-Fill* cards to automatically populate credentials and test role-specific workflows instantly.

7. Summary & Operational Conclusion

The **SVCE No-Dues Smart ERP System** represents a comprehensive, modern, institutional-grade digital transformation. By migrating away from antiquated paper clearance cards to an autonomous 6-stage sequential digital workflow backed by pure MySQL, real-time geolocation auditing, and PWA capabilities, the platform achieves:

1. **Complete Institutional Accountability:** Every approval, rejection, hold remark, and staff login is permanently recorded with exact timestamps, device names, and verified geographic locations.
2. **Zero Clearance Bottlenecks:** Students track their clearance progress in real-time from their mobile phones or laptops, receiving transparent departmental remarks and targeted re-submission channels.
3. **Tamper-Proof Certification:** External organizations, universities, and examination boards can instantly verify student clearance validity via secure QR codes linked to cryptographic hashes.
4. **Future-Ready Infrastructure:** Built on industry-standard React 18, Node.js Express, and MySQL 8.0, the architecture is easily extensible to future integrations such as automated WhatsApp alert gateways and National Academic Depository (DigiLocker) integration.

Official Institutional Release

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